

PHASE 2 — THE REFACTORING SPRINT

I Thought We Had 2,000 Tests. We Had 2 Monoliths and a Race Condition.

The honest story of refactoring 4,000 lines of AI-generated tests and the bug we found by accident.

LINES DELETED

~4,000

2 monoliths → 23 modules

TESTS RUNNING

1,200+

292 unique × 4 browsers

BUGS FOUND

1

Race — 1 line fix

AI MODELS

15

9 still free · 6 gone

THE SPLIT — VISUALIZED

2 Monoliths → 23 Modules

~4,000 lines of AI-generated spaghetti restructured into clean, domain-separated modules.

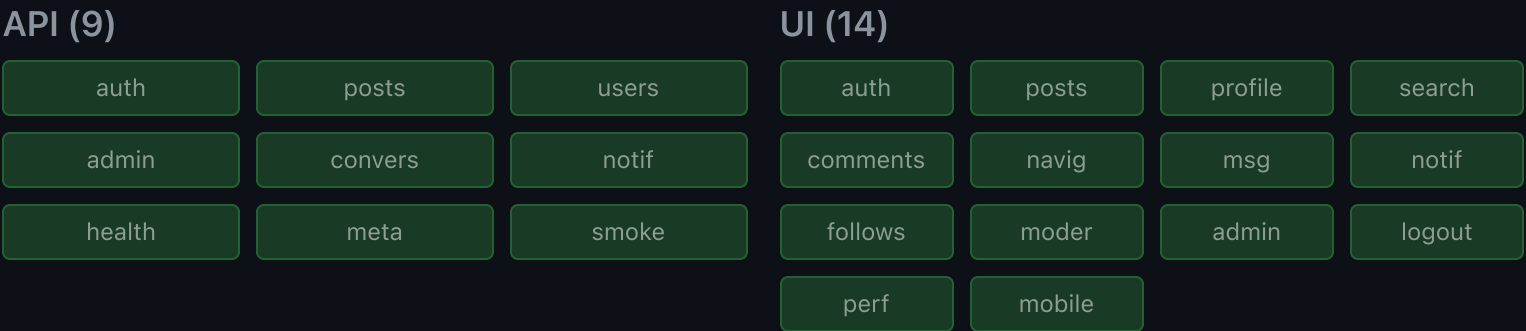
Before



3,908 lines total · 15 duplicated API blocks
No shared fixtures · Hard to maintain



After



23 modules · Shared `fixtures.ts`
CI smoke in 3 minutes · Zero monoliths

1,157 Runs · 23 Modules · 4 Browsers

Full layer diagram: from spec modules through shared infrastructure to deployment targets.



FOUND BY ACCIDENT

A Race Condition Lived in Production for 1 Month. Hidden from Sequential Tests.

Standard tests couldn't catch it — they were single-thread. Only when we split the monoliths and ran 4 parallel browser projects did the collision surface.

ROOT CAUSE

```
datetime.now()
```

Same microsecond → same JWT →
unique_violation in PostgreSQL

THE FIX

```
jti: uuid.uuid4()
```

One line in the token payload.
Every token is now unique.

BUG AGE

1 month

Lived since day 1

TRIGGER

4x browsers

Parallel runs exposed it

DETECTION

Parallel run

4 browsers exposed the race

Zero New Tests. Zero Regrets.

Adding tests is easy. Deleting ~4,000 lines, stabilizing production infra — that's where the real value was.

Phase 1 (April)

- 750+ tests built
- API coverage: 94%
- PBT coverage: 100%
- 2 monoliths ~4,000 lines
- 2 envs: Docker (local) + Render (cloud)

Phase 2 (May)

- ✓ Zero new tests written
- ✓ ~4,000 lines deleted
- ✓ 2 → 23 modular files
- ✓ CI smoke: 12 endpoints / 3 min
- ✓ Render stabilized + uptime monitor
- ✓ 1 bug found (race condition)

Lines Added vs Deleted

Phase 1

+750 tests added



Phase 2

~4,000 lines deleted



"Sometimes the best test improvement is deleting 4,000 lines and starting cleaner."

THE FREE AI MARKET IN 2026

15 Models in 2 Months. 9 Still Free. 6 Already Gone.

The treadmill is real. Don't rely on one provider.

● Active — 9 tools

OpenCode: GLM-5, M2.7, GPT-5 Nano, DeepSeek, **Nemotron 3 Super**, Big Pickle

OpenCode Zen · GitHub Copilot · Ollama qwen2.5:3b · Groq Llama 3.3 70B

ALL FREE

● Gone — 6 tools

MiniMax M2.5 (promo) · Cursor Hobby (50 req) · Cline

Gemini API (429) · qwen2.5:14b (timeout) · Continue.dev

FREE TIER ENDED



Free AI is a treadmill

Keep moving or get left behind. Maintain a stack of 4–5 free tools across different providers — and be ready to swap any week.

WHAT I LEARNED

Three Honest Takeaways

- 1 AI amplifies your prompt quality**
"Write tests" generated 4,000 lines of tech debt. A specific prompt with modular structure would have saved Phase 2 entirely.
- 2 Refactoring finds bugs that testing misses**
A race condition lived for 1 month in the API. Sequential tests couldn't catch it — only parallel execution across 4 browser projects exposed the collision.
- 3 Free AI is a treadmill, not a foundation**
15 tools in 2 months. 6 are gone. 9 remain. Diversify, rotate, never depend on a single provider.

THE CHAIN



"The biggest lesson? Deleting 4,000 lines was worth more than writing 1,000 new ones."

PHASE 3 — COMING SOON ★

This Is Only Half the Stack.

The monoliths are gone. Architecture is documented.
Now we expand across 5 stacks.



Go Backend

11 tests · 2 packages
Race detection · Table-driven tests
Error handling patterns



C# + Fuzzer

41 tests · 5 modules
Fuzzer · Properties · Schema
Race · Metamorphic



Mutation Testing

API mutations · DB mutations
Chaos testing
BUG-005 (XSS) · BUG-006

5 stacks · TypeScript · Go · C# · Python · k6 (load) | 7 bugs found | 5 architecture docs

YOUR TURN

What's the messiest test suite AI has helped you create?

Share your story in the comments. How did you clean it up —
or are you still living with the debt?

~4,000

Lines deleted

\$0

Budget

23

Modules after refactoring



Victor Ematin

AI Quality Engineering Lead

Agents under OpenCode Desktop · DeepSeek · Big
Pickle · Nemotron · OpenCode Zen